

WiFi CSI and Drone Navigation

Open-ended | roughly 6 hours

A bit of context

You built a drone that navigates without GPS using cameras and IMU. That works well outdoors and in well-lit spaces. But what about indoors, in dusty warehouses, or in environments where cameras struggle? WiFi CSI (Channel State Information) is one approach researchers have been exploring. Instead of a single signal strength number, CSI gives you detailed per-subcarrier measurements that actually reflect how the environment is shaped around you. It is an interesting idea and nobody has fully figured it out yet, which is what makes it worth exploring.

Part 1 -- Read and think

Spend some time going through research around WiFi CSI and indoor localization. A few names worth searching: SpotFi by Kotaru et al., the WiFi Sensing Survey by Ma et al., DeepFi by Wang et al. In the README of your repo, just write what you found, what makes sense to you, and where you think it breaks down for a moving drone. Keep it casual, no specific length.

Part 2 -- Try something

Grab a public CSI dataset (Widar 3.0 on IEEE Dataport is a reasonable starting point) and write a short Python script that pulls out some basic features and tries to predict location or room using a simple classifier. It does not need to work well. What matters is that you actually ran it and thought about what happened. In the README, also answer this: your dataset was probably collected with a static device, so what actually changes when the thing collecting CSI is a drone that is moving, tilting, and vibrating?

How to submit

One GitHub repo. README covers both parts. Code goes in /code. That is it.